

**INCLUSIVE MOBILE LEARNING APP FOR VISUALLY,
HEARING, PHYSICALLY AND COGNITIVELY
IMPAIRED STUDENTS (GRADE 3-5)**

25-26J-207

Project Proposal Report

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Sri Lanka Institute of Information Technology
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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

Visual impairment creates major challenges for students in accessing and understanding learning materials, especially in traditional classrooms where lessons rely heavily on visual content. In Sri Lanka, primary school children in Grades 3–5 often face difficulties when using existing educational applications because most platforms are designed for sighted users. Common accessibility tools such as screen readers and basic Text-to-Speech (TTS) solutions provide partial support, but they do not ensure a complete and independent learning experience. To address this problem, this research proposes a mobile learning application built using Flutter, designed with accessibility-first principles to support visually impaired learners.

The system integrates high-quality TTS for lesson delivery and Speech-to-Text (STT) for student responses, enabling a fully voice-driven interaction without the need for touch-based navigation. A unique feature of this solution is the use of stereo and three-dimensional (3D) spatial audio to simulate real environments, such as placing bird sounds to the left and waterfall sounds behind the listener, which helps learners build mental images through sound. Additional features include biometric login using Face ID (Facial Recognition Login) for secure access, compatibility with screen readers like TalkBack and VoiceOver, vibration feedback for real-time guidance, and an Artificial Intelligence (AI)-powered Braille document generator to produce tactile learning materials.

This approach goes beyond traditional accessibility apps by combining voice interaction, immersive audio, tactile learning, and AI-powered assistance into one integrated system. By aligning with the local curriculum and supporting Sinhala language, the application aims to increase independence, engagement, and confidence among visually impaired students. The expected outcome is an inclusive, practical, and scalable mobile platform that enhances learning opportunities for visually impaired learners in Sri Lanka and potentially in other regions.

Keywords– *Visual impairment, Text-to-Speech (TTS), Speech-to-Text (STT), spatial audio, Braille document generatio*

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
TTS	Text-to-Speech
STT	Speech-to-Text
3D	Three-Dimensional
AI	Artificial Intelligence
WCAG	Web Content Accessibility Guidelines
UI	User Interface
BRDG	Braille–Ready Document Generation
Face ID	Facial Recognition Login

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1. INTRODUCTION

1.1 BACKGROUND

Visual impairment significantly affects how students interact with learning materials. In a traditional classroom, lessons are heavily dependent on visual aids such as printed books, images, and written instructions. For students who are blind or have low vision, this creates major accessibility challenges, especially when digital learning tools are not designed with inclusive features. In Sri Lanka, many primary school students in Grades 3–5 face these barriers daily, making it difficult for them to access the same quality of education as their sighted peers.

Existing educational applications for visually impaired learners often provide partial accessibility—such as basic screen reader compatibility or simple audio playback—but they lack full voice-based navigation, interactive auditory guidance, and locally relevant content. Some require small touch interactions or complicated menu navigation, which are difficult for students who rely solely on sound and tactile feedback. This gap leaves many learners dependent on constant assistance from parents or teachers, reducing their independence.

The proposed mobile learning application, developed using Flutter, is designed to eliminate these barriers by creating a fully voice-driven and accessible learning platform. TTS to read lessons aloud in the local language and STT to record spoken answers, ensuring that every part of the learning journey can be completed hands-free. Students can register and log in using Face ID without typing, and all interactions are guided through natural, conversational voice prompts.

To make lessons more immersive, the app uses 3D directional audio and stereo spatial sound cues that help students build mental maps of learning environments. For example,

in a lesson about the jungle, bird sounds may come from one direction while a waterfall can be heard behind the listener, making the experience more vivid. The system also supports Braille–Ready Document Generation (BRDG) through an AI assistant, giving learners the option of tactile learning materials. Additional accessibility features include vibration feedback for user actions and compatibility with popular screen readers such as Talkback and Voiceover.

By combining AI–driven guidance, auditory immersion, tactile learning options, and hands–free control, this solution goes beyond traditional accessibility tools. It not only supports academic learning but also encourages independence and self–confidence in visually impaired students. This approach aligns with the local curriculum while also being adaptable for wider use beyond Sri Lanka, aiming to create a truly inclusive, engaging, and future–ready learning experience for blind and low–vision learners.

1.2 LITERATURE REVIEW

[1] B. Chatpaitoon et al., “Effect of optical text recognition (OCR) and text-to-speech — ‘CU Eye Reader’ smartphone app,” 2023.

The CU Eye Reader integrates OCR and TTS into a mobile application, enabling visually impaired individuals to read printed material using their smartphones. Its evaluation shows significant improvements in accessibility, particularly in personal reading tasks. The app demonstrates strong OCR accuracy and real-time processing, but its design is primarily focused on generic text access without curriculum alignment or child-centered interfaces. The proposed system adopts the same OCR+TTS pipeline but adapts it to Sri Lankan Grade 3–5 content. By embedding Sinhala language support, simplified navigation, and educational structuring, the new system extends CU Eye Reader’s functionality into an inclusive classroom-ready tool.

[2] “KNFB Reader — product and usability studies,” National Federation of the Blind.

The KNFB Reader is considered a pioneering OCR-to-speech product that underwent

extensive usability testing within the blind community. Its main strengths include highly accurate recognition, fast conversion, and portability. However, its commercial nature and general-purpose orientation limit its suitability for young learners in developing contexts. The proposed system incorporates KNFB Reader’s best practices—like one-tap capture and simple workflows—but enhances them with Sinhala-based child-friendly content, progress tracking, and braille export. Unlike KNFB Reader, which is document-oriented, the system integrates OCR into a gamified educational platform.

[3] J. Sánchez et al., “Navigation for the Blind through Audio-Based Virtual Environments,” 2010.

This research demonstrates how Audio-Based Virtual Environments (ABVEs) can train blind users to develop spatial awareness. By creating auditory landmarks and using spatial sound, learners can build mental maps in safe, virtual contexts. While valuable, the study focuses on mobility skill-building rather than structured education. The proposed system applies the same principle—using spatial audio cues to help visually impaired children navigate learning content. For example, sounds can be placed around the learner to indicate lesson directions or activity flow. This adapts Sánchez et al.’s navigation findings to an educational framework, supporting engagement and comprehension in Grade 3–5 classrooms.

[4] “Spatial Knowledge via Auditory Information for Blind Individuals — review,” 2022.

This review highlights how 3D audio contributes to spatial cognition and orientation for blind individuals. Evidence shows that spatial auditory cues enhance memory retention and help construct mental maps, reinforcing independence. The proposed system builds on this by embedding 3D sound into its learning modules. Rather than focusing solely on navigation, the sounds are integrated into academic storytelling, quizzes, and activity design. For instance, background sounds can simulate natural environments during a lesson, improving immersion and comprehension for young learners.

[5] G. Latif et al., “An Innovative IoT-Based AI-Powered Braille Learning System,” 2023.

Latif et al. present a hardware-driven braille learning system that connects IoT devices with AI-powered recognition to support braille education. This innovation emphasizes hands-on learning but requires specialized equipment, making it costly and less accessible in resource-limited schools. The proposed system adopts the underlying idea of AI-supported braille interaction but focuses on a software-first design. By generating printable braille PDFs and supporting braille embossing through standard printers, it reduces costs while still enabling effective braille learning. This distinction ensures broader applicability in the Sri Lankan education context.

[6] A. Shyna, “BrailleSegNet: A novel methodology for braille dataset generation and segmentation,” 2025.

BrailleSegNet introduces a deep-learning-based method for segmenting braille documents to create reliable datasets. It solves key challenges such as dot misalignment, skew correction, and noise in scanned materials. This provides a foundation for high-accuracy braille recognition and data preparation. In the proposed system, such segmentation techniques can be used to verify the accuracy of generated braille PDFs and potentially support recognition of embossed student work. By adopting these methodologies, the system ensures that braille resources for Sinhala students remain error-free and standardized for classroom use.

[7] A. Al-Salman et al., “Fly-LeNet: a deep learning framework for converting braille images to multilingual text,” 2024.

Fly-LeNet leverages deep learning to back-convert braille images into text across multiple languages, providing a reliable tool for educators and parents to evaluate student braille output. This system strengthens accessibility by allowing two-way interaction between braille and print text. The proposed system primarily focuses on forward conversion (Sinhala text to braille), but Fly-LeNet’s methodology demonstrates the value of enabling reverse recognition. Incorporating similar techniques would allow teachers to scan student braille submissions and automatically verify them, making the proposed solution more comprehensive.

[8] I. A. Elshaer et al., “Sustainable AI Solutions for Empowering Visually Impaired Students,” 2025.

Elshaer et al. stress the importance of sustainability in AI-driven assistive technology, particularly offline usability, affordability, and long-term accessibility. Their study highlights how cloud-heavy or hardware-intensive solutions exclude many learners. This aligns closely with the proposed system, which emphasizes TensorFlow Lite models for offline inference and low-bandwidth deployment in Sri Lankan schools. By prioritizing low-cost mobile devices and localized content, the proposed system addresses the sustainability concerns raised in this research, ensuring that even rural students can benefit.

[9] C. Lugaresi et al., “MediaPipe: A Framework for Building Perception Pipelines,” 2019.

MediaPipe provides efficient, real-time perception pipelines for tasks such as hand tracking, pose estimation, and object detection. It is lightweight and optimized for on-device performance, making it ideal for assistive technologies. In the proposed system, MediaPipe supports child-friendly gesture recognition to complement audio and touch navigation. Simple hand gestures, such as raising a palm or pointing, can be recognized without cloud processing. This makes the interface more inclusive, particularly for children who may struggle with fine motor skills or touchscreen interactions.

[10] S. Shokat et al., “Analysis and Evaluation of Braille to Text Conversion,” 2020.

This work evaluates multiple braille-to-text recognition techniques, comparing accuracy across different conditions such as embossing quality, lighting, and scanning resolution. The findings emphasize the importance of robust preprocessing and error correction. Although the proposed system is primarily focused on text-to-braille generation, insights from Shokat et al. are useful for ensuring high-quality output validation. They also suggest future directions for adding a braille recognition feature to support teachers in digitizing student work.

[11] Y. Vyawahare et al., “An Integrated OCR-Based Assistive System for Visually Impaired Individuals with Enhanced Accessibility,” 2025.

Vyawahare et al. present an integrated system combining OCR, TTS, and accessibility enhancements to create a single assistive application. Their design demonstrates the potential of consolidating multiple tools into one platform, which reduces complexity for users. However, the system is primarily general-purpose and lacks child-specific adaptations or curriculum-based structuring. The proposed system follows a similar integrated approach but differentiates itself by focusing on Sinhala Grade 3–5 educational content, adding gamified progress tracking, and supporting braille output for both teachers and learners.

[12] C. Albarico et al., “Development of a Mobile-Based Vision Assistant Application for Visually-Impaired Users,” 2025.

This application offers general assistance for visually impaired individuals through mobile-based object detection, text reading, and guidance features. While it improves independence, it is designed mainly for adult users in daily life contexts. In comparison, the proposed system adapts these technologies for structured education, using text reading for Sinhala lessons, integrating 3D audio for activity guidance, and minimizing cognitive load with age-appropriate UI. This shift from general life assistance to targeted educational support marks the key innovation of the new system.

[13] C. M. Bennett et al., “Understanding In-Situ Use of Commonly Available Navigation Technologies by People who are Blind,” Microsoft Research Tech Report, 2020.

Bennett et al. examine how blind users interact with commercial navigation tools in real-world settings. They find challenges such as GPS errors, signal loss, and overwhelming interfaces, which reduce trust in these systems. The proposed system avoids these pitfalls by limiting navigation tasks to controlled indoor environments such as classrooms or activity simulations. By embedding simple, reliable audio cues rather than relying on uncertain GPS signals, the system ensures predictable performance suitable for primary-school learners.

[14] K. C. Siu et al., “A Haptic and Auditory White Cane to Navigate Complex Virtual Worlds,” 2020.

This research combines haptic and auditory feedback to enhance navigation in virtual spaces. It shows that multimodal cues reduce cognitive load and increase accuracy. For the proposed system, these findings suggest the value of combining sound with haptic cues in learning environments. For example, a vibration can confirm lesson completion while spatial audio guides the next activity. Unlike Siu et al.’s cane-based solution, the system delivers these cues through a smartphone, making them accessible to children without additional hardware.

[15] M. Jain et al., “VStroll: An Audio-based Virtual Exploration to Encourage Walking for People with Visual Impairments,” 2021.

VStroll demonstrates that audio-guided exploration encourages physical activity and confidence in mobility. The study highlights how immersive audio can motivate users to engage with their surroundings. The proposed system adapts this principle to the classroom context, where spatial audio scenes can turn academic content into exploratory learning. For example, students can “walk through” a story environment using audio cues, reinforcing both mobility and comprehension.

[16] C. Zhang et al., “Hear We Are: Spatial Audio Benefits Perceptions of Turn-Taking and Social Presence in Remote Collaboration,” 2023.

Zhang et al. show that spatial audio improves turn-taking, clarity, and presence in group conversations. This finding is important for designing interactive lessons and group activities for visually impaired learners. By assigning different voices or sounds spatially, the proposed system helps students distinguish between characters, teachers, and peers during activities. This approach improves comprehension and reduces confusion in multi-voice environments.

[17] J. M. G. Calleja et al., “Use Cases and Impact of Audio-Based Virtual Exploration,” 2019.

This workshop paper reviews various scenarios where audio-only environments benefit blind users. It identifies entertainment, training, and education as promising domains for exploration. The proposed system extends this by explicitly aligning audio-based environments with Sri Lankan school curriculum content. For example, lessons can be embedded within audio-based games or virtual scenarios, transforming abstract concepts into accessible auditory experiences.

[18] L. Messi et al., “An Audio Augmented Reality Navigation System for Blind and Visually Impaired People Integrating BIM and Computer Vision,” 2025.

Messi et al. propose an advanced system integrating Building Information Models (BIM) and computer vision to provide precise navigation with spatial audio. While highly accurate, it is complex and resource-intensive. The proposed system takes a simplified approach, focusing on affordable mobile-only solutions. Instead of BIM, it uses pre-mapped classroom layouts and basic sound cues to provide guidance. This adaptation ensures feasibility for primary schools in Sri Lanka.

[19] G. Voutsakelis et al., “Development and Evaluation of a Tool for Blind Users Utilizing AI Object Detection and Haptic Feedback,” 2025.

This research integrates AI object detection with haptic feedback to assist blind users in identifying their surroundings. While effective, it risks overwhelming users with too much sensory input. The proposed system borrows this principle but applies it selectively—for instance, providing haptic confirmation only for key interactions like quiz answers. This ensures usability for young learners without cognitive overload.

[20] C. Karpodini et al., “Haptic Feedback to Overcome Barriers for Visually Impaired Users in Digital Audio Workstations,” 2025.

Karpodini et al. explore how haptic cues support complex digital tasks, showing they enhance precision and reduce barriers. For the proposed system, this reinforces the idea of using vibration feedback to guide students through lessons and confirm actions. Unlike professional workstations, the context here is simplified child learning. Hence, the haptic

design will be minimal, consistent, and playful—providing reassurance without distracting from educational goals.

1.3 RESEARCH GAP

Learning for visually impaired students is often supported through assistive technologies like screen readers, basic TTS tools, and voice-controlled navigation. While these tools provide access to information, most of them are designed for general accessibility rather than structured school education. They rarely follow the local curriculum or provide subject-based lessons for younger students in an interactive way.

Many existing applications offer voice interaction, but they are usually limited to simple commands and do not include an AI– powered learning assistant that can guide lessons, answer questions, and adjust the learning experience to suit the student’s pace. Similarly, stereo or 3D spatial audio is used in some entertainment and navigation apps, but it is not commonly applied in education to help students build mental images of a scene through sound placement.

Another limitation in current tools is the lack of integration of BRDG directly within a learning app. While standalone Braille translation software exists, there is no unified system where students can learn through voice and sound while also having lesson content automatically converted into BRDG formats. In addition, most solutions do not combine multi-modal feedback such as vibration cues and contextual auditory prompts, which can make learning more engaging and interactive.

Furthermore, features like Face ID login, compatibility with TalkBack and VoiceOver, and a smooth combination of TTS and STT are often found separately in different tools but not together in one platform tailored for school children with visual impairment.

This project addresses these gaps by creating a single Flutter-based mobile application that combines AI-powered voice guidance, stereo 3D audio storytelling, vibration feedback, contextual auditory prompts, BRDG, biometric login (Face ID), and screen-reader compatibility. It is designed specifically for Sri Lankan primary students in Grades

3–5, ensuring that accessibility and education are fully integrated into one learning experience.

Table 1: Research Paper Comparison on Visual Impairment Support and Proposed Mobile App

Future/ Capability	Paper[1]. CU Eye Reader- OCR + TTS App	Paper[2]. MediaPipe Framework for Perception Pipelines	Paper[3]. Navigation for the Blind via Audio-Based Virtual Environments	Paper[4]. IoT-Based AI- Powered Braille Learning System	Our Proposed System
Voice-based interaction for navigation. Lesson flow, and quizzes	✓	✓	✓	✗	✓
Text-to-speech (TTS) for lesson delivery	✓	✓	✓	✓	✓
Speech-to- Text(STT) for voice input	✓	✗	✓	✗	✓
Stereo spatial audio to simulate learning environments	✗	✗	✓	✗	✓
Face ID / Biometric login for secure access	✗	✗	✗	✗	✓
Compatibility with screen readers (Talkback, Voiceover)	✓	✓	✓	✓	✓
AI-powered voice assistant for guided instruction	✗	✗	✗	✗	✓
Braille-ready document generation	✗	✗	✗	✓	✓
Vibration feedback for actions	✗	✗	✗	✗	✓

1.4 RESEARCH PROBLEM

1. Why do visually impaired students encounter difficulties in Sri Lankan classrooms and digital platforms?

Many lessons depend on visual tasks such as reading, image recognition, or navigating small screen buttons. Blind or low-vision children cannot easily perform these actions, which forces them to rely heavily on external support. The proposed system introduces voice commands, vibration cues, and spatial audio guidance so students can learn independently without needing visual interaction.

2. Why are international accessibility tools not effective for Sri Lankan learners?

Most global solutions are built around English interfaces, lack Sinhala or Tamil language support, and are not designed with the local school curriculum in mind. They often do not address the real-time navigation needs of younger students. This solution is localized with Sinhala/Tamil TTS and STT and directly aligned with the Sri Lankan Grade 3–5 syllabus, making it culturally and linguistically appropriate.

3. What gaps exist in current local tools for visually impaired students?

Existing apps generally provide only audio lessons or basic screen reader support, without combining Braille document access, 3D sound, or AI-guided interaction. This limits active engagement and independent learning. The system integrates AI voice assistance, Braille-ready content, spatial audio, and tactile feedback into a single interactive platform.

4. How does missing curriculum alignment affect learning outcomes?

When lessons are not connected to the national curriculum, students cannot cover required topics or access examples familiar to their cultural and classroom context. This reduces the practical value of these tools in schools. The proposed app delivers syllabus-linked lessons, quizzes, and games designed for local classrooms, ensuring direct use in education.

5.What solution addresses these challenges?

A fully voice-driven and localized mobile learning platform with 3D audio, Braille support, quizzes, gamification, and progress tracking can provide equal learning opportunities.

This approach creates a multi-sensory, inclusive system that supports independent study while allowing teachers and parents to monitor progress.

2. OBJECTIVES

2.1 MAIN OBJECTIVES

To develop an inclusive, voice-driven mobile learning module for visually impaired students in Grades 3–5 that enables independent navigation, immersive audio-based learning, and accessible assessments using AI-powered tools, ensuring full compatibility with assistive technologies and the Sri Lankan school curriculum.

2.2 SPECIFIC OBJECTIVES

1. Enable complete voice interaction for lesson navigation, content delivery, and quizzes by integrating TTS and STT through Flutter plugins.
2. Implement stereo spatial and 3D directional audio cues to create realistic, immersive learning environments that help students understand concepts through sound positioning.
3. Integrate biometric login with Face ID to simplify secure access without typing or visual interaction.
4. Ensure compatibility with popular screen readers such as Talk-Back and Voice-Over for smooth and consistent accessibility.
5. Develop an AI-powered Braille document generator to produce curriculum-aligned tactile learning materials for offline study.

6. Provide vibration feedback and contextual auditory prompts to guide students during lessons and assessments in real time.
7. Design and implement AI-assisted voice guidance for personalized learning support and adaptive instructions.

3. METHODOLOGY

This research follows a design and development–based methodology to create a mobile learning module for visually impaired students in Grade 3–5. The design is centered on accessibility–first principles, meaning every feature is planned to work without relying on sight. The system is built Flutter for cross–platform compatibility and integrates assistive technologies like TTS, STT, 3D audio, vibration feedback, and AI–driven Braille document generation. The development process follows an iterative model, allowing continuous testing with real users at a Blind school to refine usability, accessibility, and learning experience.

3.1 System Overview

The system architecture is organized into distinct layers to guarantee accessibility, intelligent operation, and secure management of information. The interaction layer supports multiple input modes tailored to different disabilities, while the application logic delivers learning activities, games, and progress tracking. An AI and accessibility layer enhances the experience with voice assistance, Braille compatibility, and spatial audio. The data layer ensures safe storage of content and records through databases and cloud services, and the security layer provides authentication and encryption. Collectively, layers form an inclusive and dependable mobile learning environment.

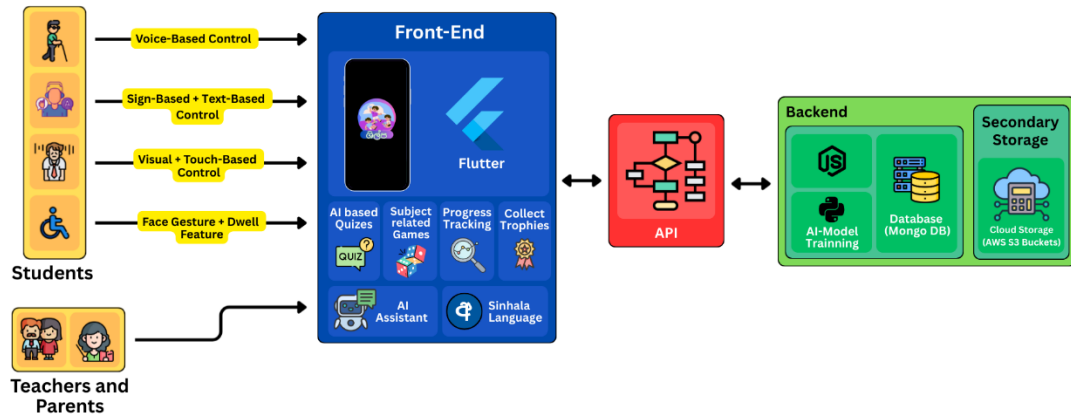


Figure 1: System overview diagram of the complete inclusive mobile learning app

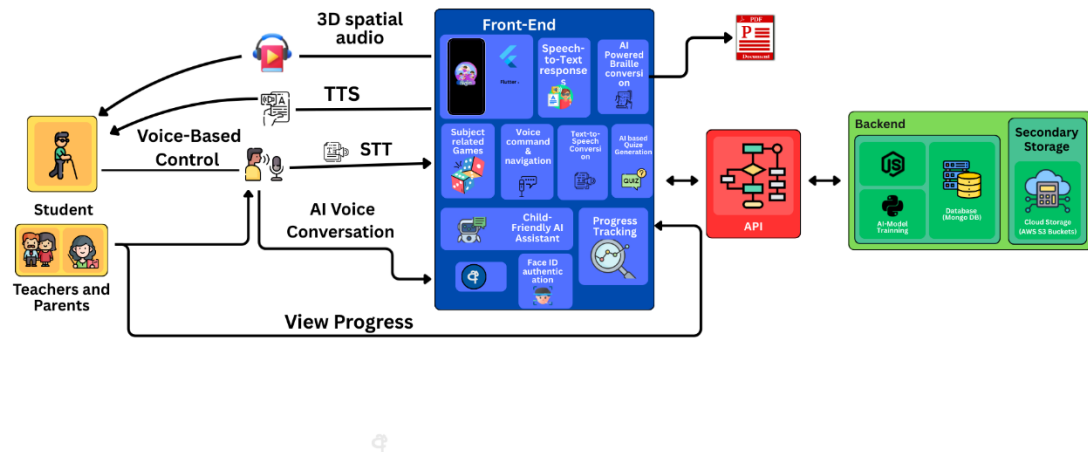


Figure 2: System architecture overview for visual impairment support

1.User Interaction Layer

This layer enables learners to interact with the system through voice commands, speech-to-text responses, 3D spatial audio, and Braille document access. Parents and teachers can also engage with the platform to monitor progress and provide support.

2.Application Logic Layer

The application core, built on Flutter, manages subject-based games, voice navigation, quiz generation, and real-time progress tracking. It also integrates Sinhala language support and ensures smooth learning activities through structured logic.

3.AI & Accessibility Layer

Advanced features such as the child-friendly AI assistant, speech recognition (STT), text-to-speech conversion (TTS), Braille document generation, and spatial audio create a personalized and inclusive learning experience for students with disabilities.

4.Database & Storage Layer

User profiles, quiz data, Braille files, and multimedia lessons are stored securely in a MongoDB database, with scalable cloud storage (AWS S3) providing backup and easy access to large resources.

5.Security Layer

Face ID authentication, encrypted storage, and controlled sessions ensure privacy, safety, and secure handling of sensitive student and teacher information.

3.2 Development Tools and Technologies

The system is developed using Flutter as the frontend framework, which allows cross-platform mobile development for both Android and iOS devices with a single codebase. The core programming language is Dart, which ensures smooth UI rendering and responsive performance. To make the app accessible, Google TalkBack and Apple VoiceOver APIs are integrated, so visually impaired students can interact easily with the interface. For voice-based features, the system uses Hugging Face models to provide high-quality TTS and STT functionalities. To create immersive learning experiences, a 3D Audio Engine such as OpenAL or the Unity Audio SDK is connected to Flutter through plugins, allowing spatial audio effects in lessons.

All learning content, quiz results, and user profiles are securely stored in Firebase Cloud Firestore, while Firebase Authentication with Face ID support ensures safe and simple login for students. For AI-powered features such as Braille document generation and the AI learning assistant, a Python-based backend is built using FastAPI, which connects with the Flutter frontend via APIs. Version control and team collaboration are managed through GitHub, ensuring proper code tracking, updates, and collaborative development across the project team.

3.3 Functional Modules & Workflows

1. **Voice-Based Navigation Module** – Handles all app interactions via voice commands (lesson selection, navigation, quizzes).
2. **Audio Lesson Module** – Uses TTS to read content aloud with 3D spatial sound for immersive learning.
3. **Quiz & Feedback Module** – AI-generated voice quizzes with instant spoken feedback and vibration alerts.
4. **Braille Document Generator** – Converts lesson text into Braille-ready files for printing or embossing.
5. **Authentication & Security Module** – Uses Face ID login and secure session management.

3.4 Data Collection

Data was collected from the School for the Blind in Tangalle to better understand the specific learning needs of visually impaired students. Information was gathered through interviews, observation sessions, and small pilot testing activities with both students and teachers. The focus was on identifying learning preferences such as audio, tactile, or vibration feedback, challenges faced when using existing learning apps, preferred voice styles, speeds, and languages, as well as the usefulness of environmental audio cues. In addition, feedback was collected on the usability of Braille-generated documents. This data provided valuable insights to design a more inclusive and user-friendly mobile learning application.

Security and Session Handling

- **Authentication:** Face ID for login to prevent unauthorized access.
- **Secure Storage:** Firebase with encrypted storage for sensitive data.
- **Session Timeout:** Automatic logout after inactivity to maintain privacy.
- **Data Protection:** Compliance with local and international accessibility data standards.

3.5 Expected Outcomes

- A fully voice-controlled learning app that visually impaired students can use independently.
- 3D directional audio-based lessons for immersive and engaging learning.
- AI-assisted Braille document generator to provide tactile study materials.
- Improved accessibility with vibration feedback, contextual prompts, and screen reader compatibility.
- Positive learning impact on students' engagement, confidence, and independence.

3.6PROJECT REQUIREMENTS

3.6.1 Functional Requirements

These describe the features the system must provide for visually impaired students.

1. Voice-Based Navigation

- The app must allow all navigation, lesson selection, and quiz interactions using voice commands.

2. Text-to-Speech (TTS)

- The system must read out all lesson content aloud in the local language using high-quality TTS.

3. Speech-to-Text (STT)

- The app must convert students' spoken answers into text for quizzes, forms, and chatbot interaction.

4. Spatial Audio Learning Environment

- The app must simulate real-world scenes with 3D directional audio (e.g., jungle, classroom, city) to enhance learning.

5. Face ID Login

- Students must be able to log in without typing by using biometric authentication (Face ID).

6. Screen Reader Compatibility

- The app must be compatible with screen readers such as TalkBack (Android) and VoiceOver (iOS).

7. Braille-Ready Document Generation

- The system must generate Braille-ready files from lesson content using AI assistance.

8. AI-Powered Voice Assistant

- The app must have an AI voice assistant to guide students step-by-step during lessons, navigation, and quizzes.

9. Vibration feedback

- The device must provide vibration cues for important actions (e.g., quiz submission, error alerts).

10. Progress Tracking

- The app must store each student's completed lessons, quiz scores, and progress for review by parents/teachers.

3.6.2 Non-Functional Requirements

1. Accessibility
2. Performance
3. Cross-Platform
4. Data Security & Privacy
5. Scalability
6. Local Language Support

3.6.3. User Requirements

The user requirements for visual impairment support focus on providing a fully voice-driven and accessible learning experience. Visually impaired students need secure login through Face ID, navigation using voice commands, listening to audio lessons, experiencing 3D spatial sound, and completing voice-based quizzes with support from an AI assistant. They also require vibration feedback, Braille document access, and compatibility with screen readers to ensure independence. Teachers need functions to manage content and track student progress, while parents require access to progress reports. The AI system supports these needs by processing speech commands, generating Braille-ready documents, and creating immersive 3D audio environments, making the platform inclusive and effective for learners.

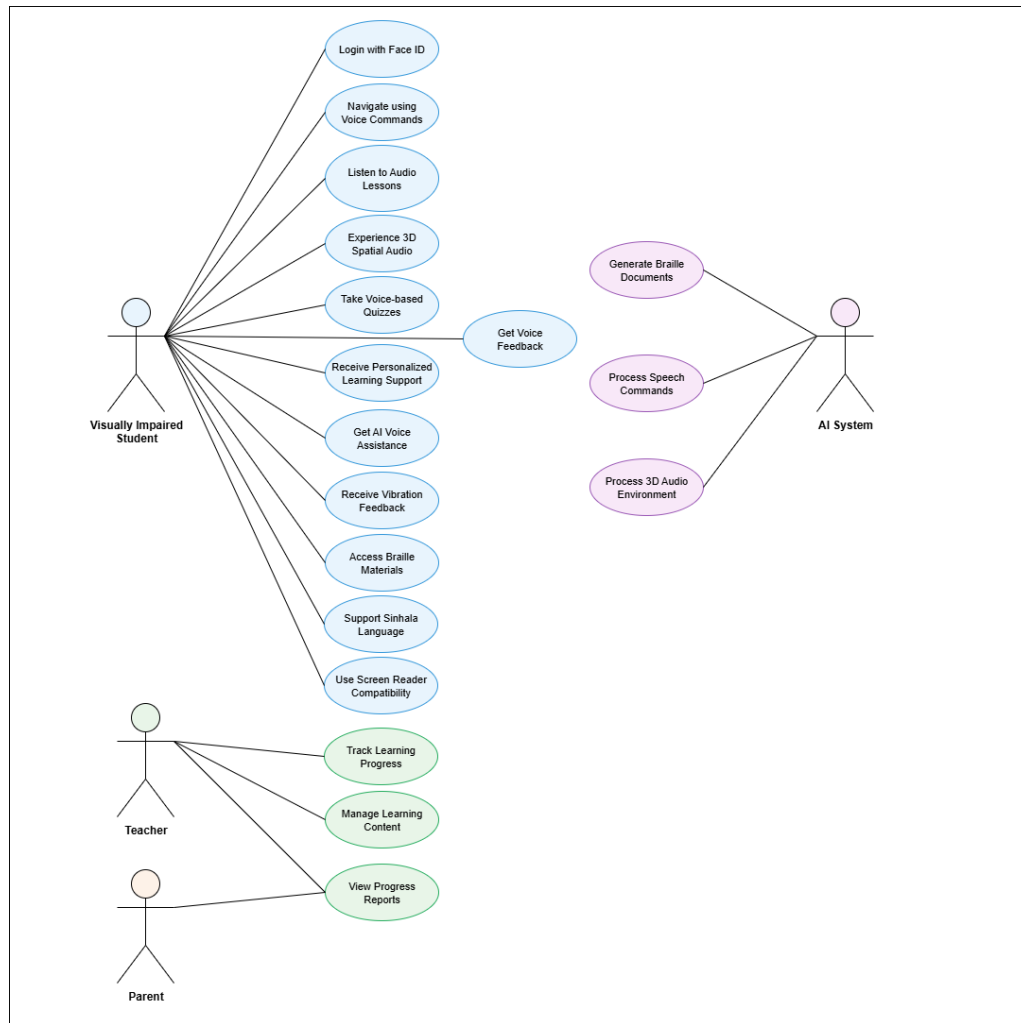


Figure 3: Use case diagram for hearing impairment support

4. BUDGET AND BUDGET JUSTIFICATION

Table II. Explains Estimated costs for assist visual impaired students support.

Table II: Budget Plan for Inclusive Mobile Learning App for visual impairment support

Purpose	Estimate cost (LKR)
Travel to special schools, interviews with teachers & students	6000
Voice Narration Recording lessons	12000
Internet cost	8000
Braille Documentation Printing	5000
Screen Reader Software	6000
Text to Speech integration Open-Source API	5000
Total	42000

5. COMMERCIALIZATION ASPECTS OF THE PRODUCT

The Visual Impairment Support module presents clear market opportunities because many Sri Lankan students with blindness or low vision lack access to suitable digital learning platforms. Most existing apps are designed for sighted learners, while international assistive tools often ignore Sinhala or Tamil and are not linked to the national curriculum. By offering voice-based navigation, Braille document generation, vibration feedback, and 3D spatial audio, this system introduces a localized, curriculum-friendly solution that directly addresses these unmet needs.

Commercial deployment can target special education schools, inclusive classrooms, and home learning environments. Features such as Braille-ready files also add value for institutions with embossing facilities, widening adoption potential.

Looking beyond Sri Lanka, the product can be extended to regional and international contexts where similar gaps exist. By localizing language and adapting to different

curriculums, the solution can be scaled to neighboring countries, especially within South Asia. Its unique combination of AI voice assistance, vibration, and curriculum alignment positions it as an attractive option for government school supporting educational impact and commercial viability.

6. WORK BREAKDOWN STRUCTURE

Task \ Time	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6	Month 7	Month 8
Requirement analysis & Research								
Data gathering, Proposal Doc & Presentation								
Prototype Design								
App Development								
AI related Developments								
Integration & Testing								
Validation & Feedback								
Final Presentation								

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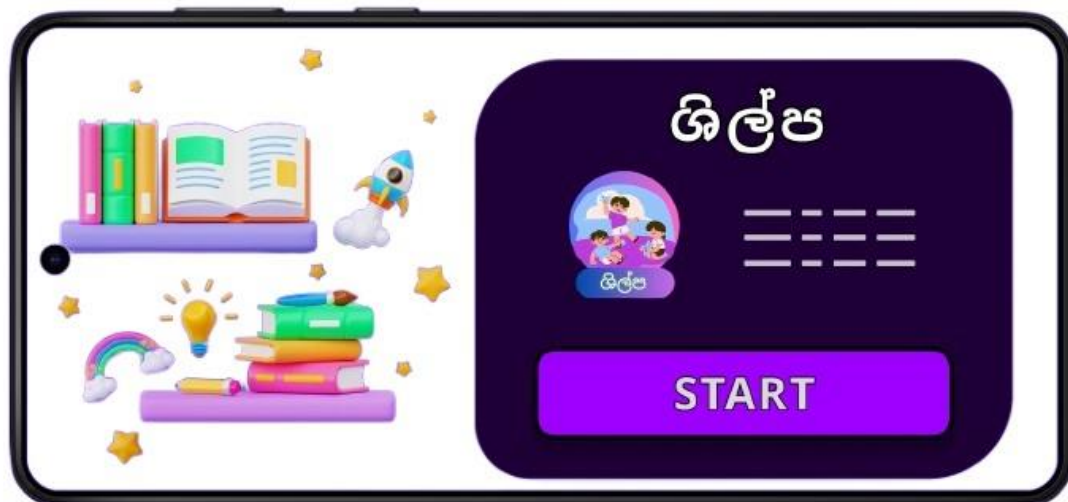
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8. APPENDICES

Appendix A: Mobile App logo



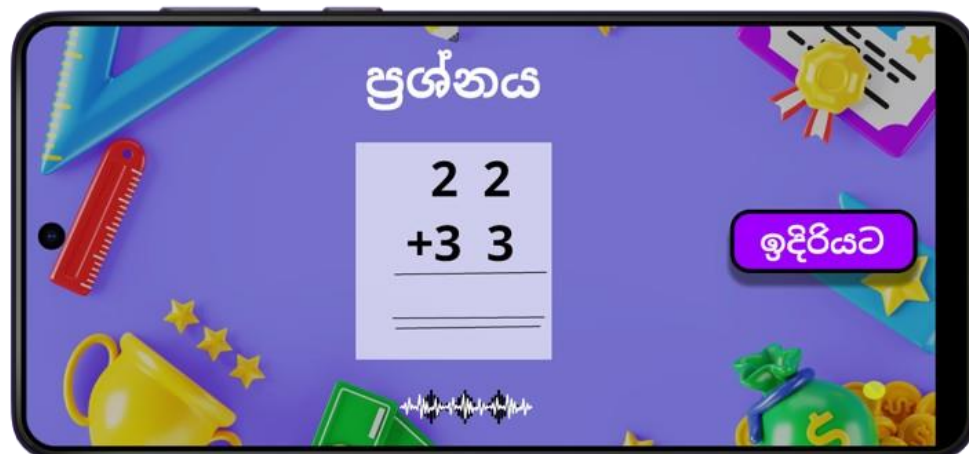
Appendix B: Application landing page



Appendix C: Appendix C: User Interface for Disability Type Selection



Appendix D: Audio-Based Math Quiz for Visually Impaired Learners





Appendix E: Data Collection at NIE Braille Press



Appendix F : Plagiarism Report

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